

**ATTACHMENT 17 – LETTER FROM BECHTEL / PARSONS BRINCKERHOFF
(AUTHORIZED REPRESENTATIVE) TO MODERN CONTINENTAL
COMPANY DATED JANUARY 7, 2000
(3 pages)**



Massachusetts Turnpike Authority

Central Artery/Tunnel Project

January 7, 2000

Letter No.: C09B2-0499
File No.: CN-9.09B2.02
CN-9.09B2.49.01.001

Mr. Robert Coutts
Project Manager
Modern Continental Companies, Inc.
600 Memorial Drive
Cambridge, MA 02210

Subject: Central Artery (I-93)/Tunnel (I-90) Project
Contract No. 99119-C09B2
I-90 Seaport Access Tunnel Finishes
Deficiency Report (DR) No. 001 - Corrective Action

Dear Mr. Coutts:

Attached for your action is the Project's corrective action for Deficiency Report Number 001 (DR-001), the failure of adhesive anchors in the HOV ceiling. Please incorporate these corrections into the work.

If you have any questions or concerns, please contact me at 350-7594 ext 56

Sincerely,

BECHTEL/PARSONS BRINCKERHOFF



Robert E. Steffy /
Resident Engineer
Authorized Representative of
the Project Director

RES/DEH/dh

Attachment: DR-001 Corrective Action (2 pages)

cc: M. Charney
G. Baxter

January 6, 2000
C09B2 Tunnel Finishes
DR-001 Adhesive Anchors Failures

The DR-001 proposed fix for the failed anchor bolts, as submitted, is rejected.

The repair/replacement procedure for the failed anchors is as follows:

1. Remove the failed anchors without damaging the hole or adjacent concrete.
2. Use a carbide bit with hammer drill to re-drill/ream the hole to create a new clean surface. There shall be no foreign matter remaining in the hole. Do not exceed manufacturer's tolerance for hole diameter. If hole diameter exceeds tolerance then change bits, hammer drill a larger diameter hole and install the next larger size anchor and maintain 5 inch hole depth.
3. Follow manufacturer's procedures to clean and inspect the hole. Begin grouting from the bottom of the hole working towards the surface or opening of the hole. Install a clean, new anchor and tension test the anchor using the governing Allowable Tension value from the catalog. For a 5/8" dia. Powerfast adhesive anchor the limiting Allowable Tension value is the adhesive bond strength, 6350 lbs.
4. Install hanger attachment plate and torque anchor bolts to manufacturer's instructions.

Because of evidence concerning installation problems, i.e. dust in the hole and insufficient adhesive in the hole, the capacity of all of the other anchor bolts is in question and will need to be addressed.

There are two options to resolve this problem.

1. Remove all of the anchors and follow the DR procedures to install new bolts. Perform tension test on each bolt using 6350 lbs, reinstall hanger and torque bolts .
2. Retest all of the anchors using the manufacturer's catalog and the governing Allowable Tension for adhesive bond strength - 6350 lbs .